Fareed Dubiure

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Education

Cornell University, College of Engineering - Ithaca, NY
Bachelor of Science in Biomedical Engineering, Expected May 2028
Molecular & Cellular Systems Concentration

Saint James' Seminary and Senior High School - Sunyani, Ghana
West African Senior School Certificate Examination (WASSCE), November 2023.
Earned 8 As (highest distinction) in both private and public examinations

Relevant Coursework: Molecular & Cellular Biology, General Chemistry, Organic Chemistry Biomolecular Thermodynamics, Investigative Biology, Biomedical Transport Phenomena, Statics & Mechanics of Solids, Physiology of Human Health & diseases, Calculus, Differential Equations, Linear Algebra, Electromagnetism.

Research Experience & Academic Projects

Analysis of Gene Expression Progression in Early-Stage Breast Cancer
Independent Bioinformatics Research Project - November 2025

- Conducted full differential gene expression analysis using the GSE16873 microarray dataset to investigate early breast cancer progression
- Identified ACTA2 as a consistently downregulated gene across disease stages, suggesting early myoepithelial and structural deterioration
- Performed data preprocessing, normalization, and statistical analysis using Python
- Wrote a complete scientific-style report including abstract, methods, results, and discussion supported by primary literature

Poster Presentation
Investigative Biology Course - November 2025

- Designed and conducted an experimental study investigating the effect of fruit waste on algal growth
- Collected and analyzed biological data to assess growth trends and experimental outcomes
- Presented findings through a scientific poster to peers and faculty, emphasizing experimental design and interpretation

Biology Laboratory Assistant
Saint James' Seminary and Senior High School - June 2022 – November 2023

- Prepared reagents, solutions, and laboratory equipment for instructional biology experiments
- Maintained laboratory safety standards and ensured proper cleanup and storage of materials
- Assisted and supervised junior students during laboratory exercises, verifying procedural accuracy

Technical Skills

***Laboratory Techniques*:** Microscopy, Sterile technique, Pipetting, Titrations, PCR, ELISA, Gel electrophoresis, Spectrophotometry, Solution preparation.

***Data Analysis & Computing*:** Python, R (basic), Biological data interpretation

Languages: English (Native), French (Intermediate, ~A2)

Teaching & Academic Support

***Math Tutor*,** April 2024 – August 2024
Afex Hub - Ghana

- Tutored students in SAT mathematics, developing customized practice materials
- Reviewed assessments and clarified conceptual misunderstandings through targeted instruction

Campus & Professional Experience

Student Program Assistant, August 2025 – Present
Tatkon Center, Cornell University

- Support daily operations of a student resource center through front desk management
- Assist with planning and implementation of workshops and peer engagement initiatives
- Provide guidance to students navigating academic and campus resources

***Museum Operations Assistant*,** June 2025 – August 2025
Johnson Museum of Art, Cornell University

- Supported museum operations and visitor engagement
- Assisted with floor monitoring and policy adherence

Affiliations & Memberships

- American Society for Biochemistry and Molecular Biology (ASBMB)

- Food Recovery Network
- The Wildlife Society